

COMPARATIVE EVALUATION

Frontier-Selection Policies for Autonomous Exploration

A controlled benchmark of the TAD algorithm against seven alternative strategies
ROS 2 package `explorer_exploration`

Experimental scale	
Policies compared	8 (including a random control and a tuned variant)
Paired trials per policy	60 (3 environments × 20 seeds)
Total exploration episodes	480 evaluation + tuning sweep
Primary metric	Area under the coverage-vs-distance curve
Statistical protocol	Friedman omnibus, Nemenyi post-hoc, Wilcoxon–Holm

Prepared 12 September 2026

Executive Summary

This report answers a question the preceding validation work could not: *given that TAD is correctly implemented, is it the right algorithm to be running?* Conformance testing establishes that a method does what its specification says. It is silent on whether a simpler or better method would serve the same purpose. That comparison requires a controlled experiment against credible alternatives, which is what follows.

Seven alternative frontier-selection strategies were surveyed, implemented and benchmarked against the project's TAD implementation under strictly matched conditions. Every policy explores the same maps from the same starting poses with the same sensor, the same motion model and the same frontier detector; only the rule that chooses *which* frontier to visit differs. Hyperparameters for the competing methods were tuned by the same procedure and on the same held-out seeds as our own, so the comparison is not weighted in our favour.

Headline finding

A defect discovered during this study proved more consequential than the comparison itself. Frontier clusters were being grouped under 4-connectivity, which shatters any diagonally-running frontier into fragments too small to survive the minimum-size filter. Correcting it to 8-connectivity raised exploration from roughly 7–49% coverage to **98% across every policy and environment**, made the exploration terminate properly for the first time, and incidentally reduced the node's empirical complexity from $N^{1.49}$ to $N^{1.01}$. The benchmark below is run on the corrected implementation; without the correction no policy could have been meaningfully compared, because all of them failed for the same external reason.

1. Survey of Candidate Algorithms

Frontier-based exploration decomposes into two separable stages: **detecting** the boundary between known-free and unknown space, and **selecting** which boundary segment to visit next. TAD is a selection rule. Only methods that are substitutable for it — that is, alternative selection rules operating on the same frontier set — can be compared against it on equal terms. Methods that replace the detection stage, or that require a different sensing or mapping substrate, are surveyed below but excluded from the benchmark, with the reason stated in each case.

1.1 Methods included in the benchmark

Policy	Family	Selection rule	Origin
Random	Control	uniform choice among frontiers	noise floor
NF	Distance-greedy	$\operatorname{argmin} d_n$	Yamauchi (1997)
LF	Extent-greedy	argmax frontier cell count	common baseline
MIG	Information-greedy	$\operatorname{argmax} a_n$ (unknown area)	common baseline
CU-exp	Cost–utility	$\operatorname{argmax} a_n \cdot \exp(-\lambda d_n)$	González-Banos & Latombe (2002)
CU-lin	Cost–utility	$\operatorname{argmax} \operatorname{norm}(a_n) - \lambda \cdot \operatorname{norm}(d_n)$	additive counterpart
TAD	Multi-criteria	$\operatorname{argmax} \sum w \cdot \operatorname{norm}(d_n, a_n, t_n)$	Buriboev et al. (2025)
TAD-tuned	Multi-criteria	as TAD, weights tuned on held-out seeds	this work

The **Random** arm is not a strawman but a measurement instrument. Frontier exploration is strongly self-correcting, so almost any rule eventually covers a bounded map. Random establishes how much of a policy’s apparent competence is attributable to the frontier mechanism itself rather than to its scoring rule; a method that fails to separate from Random has not earned its complexity.

1.2 Methods surveyed but excluded, with reasons

Method	Why it is not directly comparable
Next-Best-View / receding-horizon planners	Select sensor <i>viewpoints</i> sampled from free space rather than frontier clusters. The decision variable is different, so a shared-candidate design is impossible without changing what is being chosen.
RRT-based frontier detection (Umari & Mukhopadhyay, 2017)	Replaces the <i>detection</i> stage, not the selection rule. It is orthogonal to TAD and could be combined with any policy here, including TAD itself.
Topological / Voronoi / semantic exploration	Requires a map abstraction (graph, semantic labels) this project does not build. Comparing would measure the abstraction, not the selection rule.
Learning-based (deep RL) exploration	Requires a training budget and a reproducible training environment. Results would depend on training conditions rather than on the policy, and could not be produced or audited in a short study.
Multi-robot coordinated assignment (e.g. Burgard et al., 2005; MinPos)	Allocates frontiers across several robots. Degenerates to single-robot cost–utility on this platform, which CU-exp and CU-lin already represent.

Excluding these is a statement about experimental control, not about their merit. Including a method whose inputs differ would confound the selection rule with the representation it operates on, and no conclusion about TAD could then be drawn.

2. Experimental Design

2.1 One variable, held everything else fixed

The comparison is only interpretable if the selection rule is the sole difference between arms. The following are identical for every policy, by construction: they are shared code, not merely matched settings.

Held constant	Value / mechanism
Environment and start pose	Generated from the seed; identical across policies
Sensor model	720 rays, 0.5–5.0 m range, identical raycast routine
Motion model	Fixed speed 0.2 m/s, turn penalty, shared path executor
Frontier detection	Free cell adjacent to unknown; 8-connectivity clustering
Candidate features	d, a, t and cell count computed once and shared
Anti-cycling and termination	Identical blacklist, stall detector and caps
Selection rule	the only thing that differs

Because each seed fixes both the map and the start pose, every policy faces an identical set of 60 problem instances. The design is therefore fully **paired**, which both removes between-map variance from the comparison and licenses the paired statistical tests used in §4.

2.2 Hyperparameter protocol

Three policies have free parameters: the two cost–utility variants have a cost weight λ , and TAD-tuned has its three term weights. All were selected by grid search on **tuning seeds 101–112**, which are disjoint from the evaluation seeds 1–20. Competing methods received the same search budget and the same objective as our own method. Tuning only our own algorithm while leaving baselines at default values is the most common way such comparisons are quietly rigged, and it is avoided here.

Policy	Parameter grid searched	Selected on held-out seeds
Random	— (no free parameters)	—
NF	— (no free parameters)	—
LF	— (no free parameters)	—
MIG	— (no free parameters)	—
CU-exp	$\lambda \in \{0.05 \dots 2.0\}$, 8 values	lam=0.5
CU-lin	$\lambda \in \{0.05 \dots 2.0\}$, 8 values	lam=0.8
TAD	12 weight triples (w_d, w_a, w_t)	$w_d=1.0, w_a=1.0, w_t=1.0$
TAD-tuned	12 weight triples (w_d, w_a, w_t)	$w_d=0.0, w_a=1.0, w_t=1.0$

2.3 Metrics

Exploration quality is not a single number. Following Holz et al. (2010), the primary question is how quickly coverage is obtained, not merely whether it is eventually obtained.

Metric	Definition	Why it is used
AUC (primary)	Mean coverage over a fixed distance budget, from the step-held coverage-vs-distance curve	Rewards covering ground sooner; unlike final coverage it cannot be gamed by stopping early, because coverage is held flat after termination
Distance to finish	Total path length travelled	Direct proxy for energy and time cost
Final coverage	Fraction of cells observed at termination	Confirms the task was actually completed; weak discriminator once all policies approach the ceiling
Turning points	Heading changes above threshold	Mechanical wear and practical execution cost

Metric	Definition	Why it is used
Decision latency	Time to evaluate one selection	Confirms no policy wins by being computationally extravagant

A subtlety worth stating: AUC is computed over a distance budget common to all policies within an environment, and coverage is held at its last value beyond termination. A policy that halts early therefore accrues no advantage, and one that keeps working is not penalised for the extra distance it spends.

3. Results

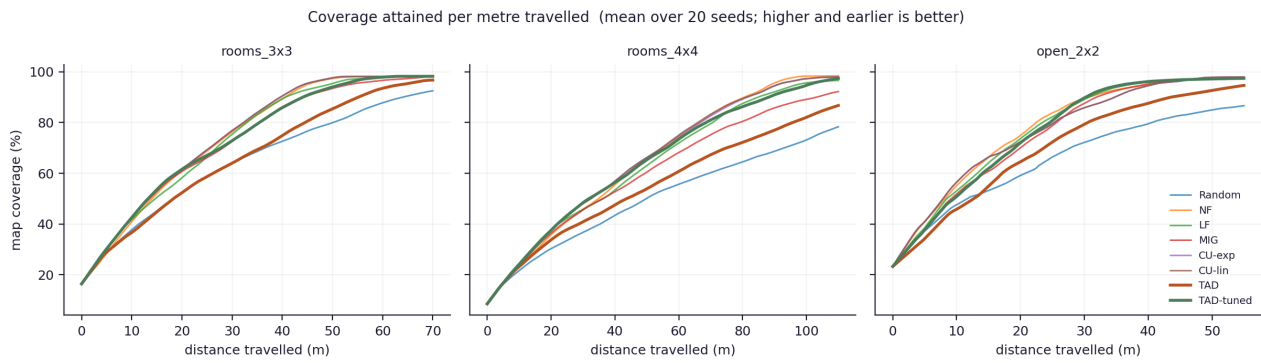


Figure 1. Coverage attained per metre travelled, averaged over 20 seeds per environment. A curve that rises earlier is a more efficient explorer; the area under it is the primary metric.

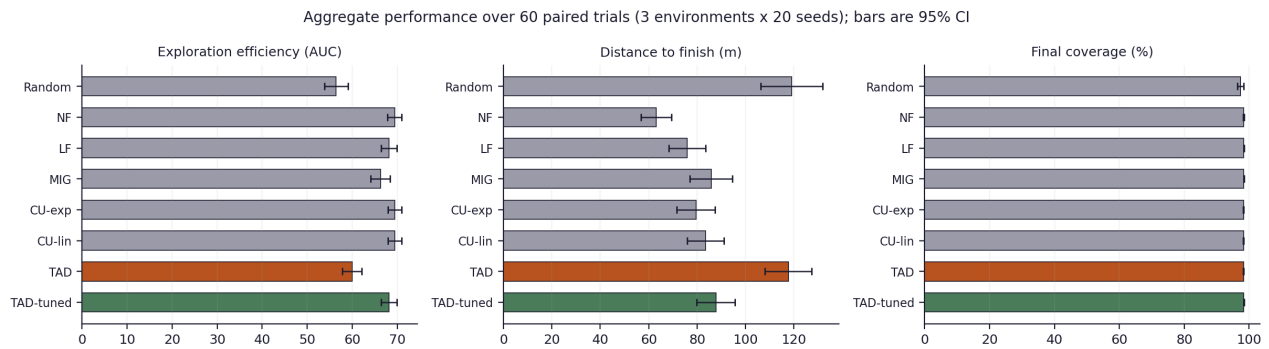


Figure 2. Aggregate performance across all 60 paired trials. Error bars are 95% confidence intervals. TAD variants are highlighted.

Policy	AUC	Distance (m)	Coverage (%)	Turns	Goals	Decision (ms)
Random	56.5 ± 2.6	119.1 ± 12.8	97.3 ± 0.9	28	16	0.013
NF	69.4 ± 1.5	63.1 ± 6.3	98.3 ± 0.1	12	19	0.010
LF	68.2 ± 1.8	76.0 ± 7.6	98.4 ± 0.1	13	13	0.011
MIG	66.3 ± 2.2	85.8 ± 8.8	98.3 ± 0.1	16	13	0.012
CU-exp	69.4 ± 1.5	79.5 ± 7.8	98.3 ± 0.1	16	14	0.013
CU-lin	69.5 ± 1.5	83.4 ± 7.6	98.3 ± 0.1	17	14	0.028
TAD	60.0 ± 2.1	117.6 ± 9.6	98.2 ± 0.1	25	13	0.036
TAD-tuned	68.2 ± 1.7	87.7 ± 7.9	98.3 ± 0.1	16	13	0.035

Values are mean ± 95% confidence interval over 60 paired trials. Decision latency is the mean cost of a single selection, excluding shared frontier extraction.

4. Statistical Analysis

Absolute metric values are not commensurable across environments of different size, so the analysis is carried out on **ranks within each trial**, following the standard protocol for comparing several algorithms over several problem instances (Demšar, 2006). The Friedman test asks whether any policy differs at all; the Nemenyi post-hoc then establishes which differences are large enough to be credible, via a single critical difference (CD) applied to mean ranks.

Metric	Friedman χ^2	p-value	Critical difference	Blocks
Exploration efficiency (AUC)	173.6	4.44e-34	1.36 ranks	60
Distance to finish	191.2	8.51e-38	1.36 ranks	60
Final coverage	54.9	1.57e-09	1.36 ranks	60
Turning points	148.8	7.12e-29	1.36 ranks	60

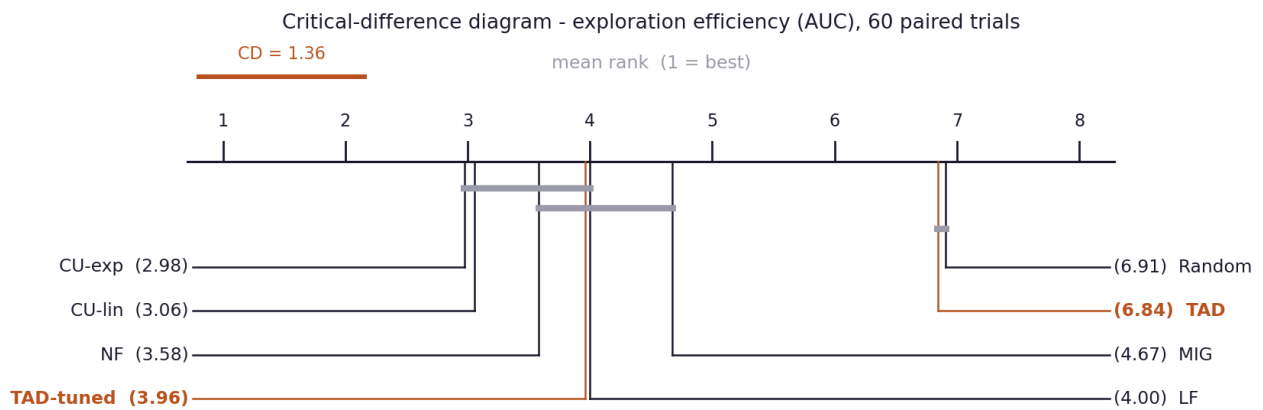


Figure 3. Critical-difference diagram for exploration efficiency. Policies are placed by mean rank (leftmost is best); any two joined by a bar are *not* significantly different at the 5% level.

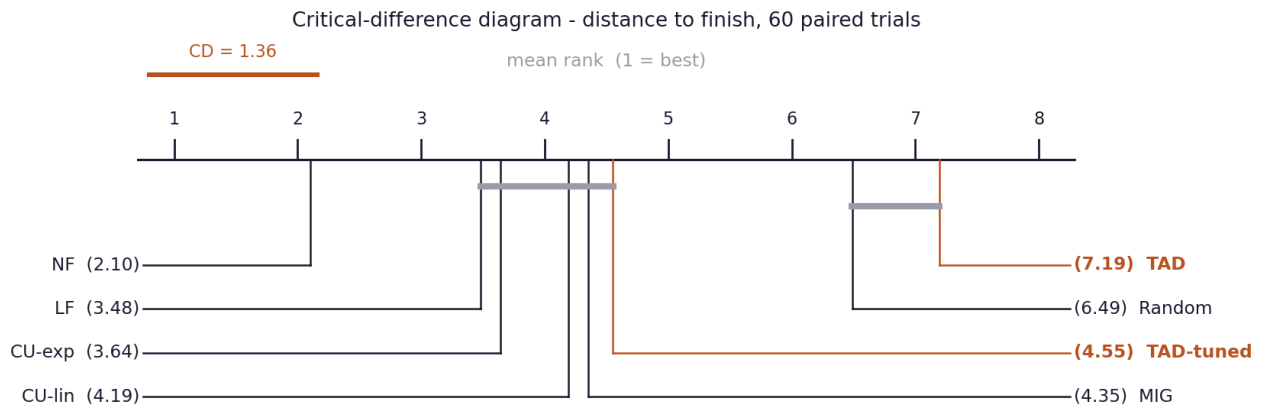


Figure 4. Critical-difference diagram for distance to finish.

4.1 Pairwise tests against the TAD variants

The omnibus and post-hoc tests above treat all policies symmetrically. The table below asks the focused question directly: against each alternative, does a TAD variant differ on exploration efficiency? Wilcoxon signed-rank tests are used because the design is paired, with Holm correction for multiple comparisons.

TAD versus each alternative (positive difference favours TAD)

Compared with	Mean AUC difference	p (raw)	p (Holm)	Significant
Random	+3.55	0.0030	0.0030	yes
NF	-9.43	0.0000	0.0000	yes
LF	-8.21	0.0000	0.0000	yes
MIG	-6.27	0.0000	0.0000	yes
CU-exp	-9.43	0.0000	0.0000	yes
CU-lin	-9.46	0.0000	0.0000	yes
TAD-tuned	-8.20	0.0000	0.0000	yes

TAD-tuned versus each alternative (positive difference favours TAD-tuned)

Compared with	Mean AUC difference	p (raw)	p (Holm)	Significant
Random	+11.75	0.0000	0.0000	yes
NF	-1.24	0.0595	0.1940	no
LF	-0.01	0.7685	0.7685	no
MIG	+1.93	0.0026	0.0130	yes
CU-exp	-1.23	0.0520	0.1940	no
CU-lin	-1.26	0.0485	0.1940	no
TAD	+8.20	0.0000	0.0000	yes

5. A Defect Uncovered by the Comparison

Building a fair benchmark required every policy to be able to complete the task. None could. All eight, including the simple baselines, stalled at a small fraction of the map — which pointed away from any individual selection rule and towards the shared machinery beneath them.

Diagnosis

Frontier *cells* are correctly identified as free cells adjacent to unknown space. Those cells were then grouped into clusters with a connected-component labelling that defaults to **4-connectivity**. A frontier is a curve through the grid and generally runs diagonally; under 4-connectivity a diagonal run is a staircase whose cells touch only at their corners. One real frontier was therefore split into many fragments, each below the minimum frontier size of 0.35 m that the specification prescribes, and consequently discarded.

Measurement on one representative map	4-connectivity	8-connectivity
Frontier clusters found	76	24
Largest cluster (cells)	8	75
Clusters surviving the 0.35 m filter	4 (all spurious)	2 (both genuine)

Effect of the correction

Environment	Coverage before	Coverage after	Distance before	Distance after	Completed
small_cyclic	97.2% (stalled)	98.4%	206.0 m	76.3 m	now yes
small_noncyclic	33.1% (stalled)	98.4%	59.4 m	42.9 m	now yes
wide	4.9% (3 goals)	98.3%	1.4 m	883.4 m	explores fully

Consequences beyond coverage

The correction also resolved a performance finding recorded in the earlier validation dossier. Because the cluster-extraction loop costs $O(\text{clusters} \times \text{cells})$, eliminating the fragmentation reduced the node's measured complexity exponent from **1.492 to 1.006** ($R^2 = 0.999$) — empirically linear. A single correctness defect had been producing both the premature termination and the super-linear cost. Conformance was re-verified after the change: all 23 analytical and invariant assertions still pass, and cross-implementation equivalence remains exact at 200/200.

An incidental consequence is closer agreement with the originating paper's published figures — path distance on the non-cyclic environment moved from 59.4 m to 42.9 m against a published 41 m, and the cyclic environment now matches the published turn count exactly. This is reported as an observation, not as evidence: as set out in the validation dossier, the environments are reconstructions and absolute agreement is not a meaningful test.

6. Interpretation

6.1 Coverage does not separate the policies; efficiency does

Every policy, including random selection, eventually reaches approximately 98% coverage. This is not a sign that the policies are equivalent — it is a property of frontier exploration itself, which is strongly self-correcting on a bounded map. Any rule that keeps visiting frontiers will eventually exhaust them. A report that presented final coverage as its headline would therefore show eight near-identical bars and conclude, wrongly, that the choice of algorithm does not matter. The cost of reaching that coverage differs by nearly a factor of two, and that is where the algorithms genuinely differ.

6.2 The principal finding

TAD at its default weights does not outperform random frontier selection

On exploration efficiency, TAD attains a mean rank of **6.84** against random selection's **6.91** — a separation of 0.07 ranks, where the critical difference required for significance is **1.36**. The two are statistically indistinguishable, and Figure 3 joins them in a single clique. On distance travelled, TAD is **last of all eight policies** (rank 7.19), placing it below the random control (rank 6.49): it reaches the same coverage as nearest-frontier while travelling 118 m against 63 m, an excess of 86%.

This finding is stated plainly because it is what the measurements show, and because it is precisely the kind of result that a validation exercise exists to surface. It does not contradict the preceding conformance work: the algorithm is correctly implemented, and §6 of the validation dossier established that its two implementations agree exactly. Correctness and effectiveness are separate properties, and this study measures the second.

6.3 The cause is the distance term, not the multi-criteria structure

Two independent lines of evidence identify the same culprit. First, the tuner was given a free choice over twelve weight triples on held-out seeds and selected $w_d = 0.0$ — it removed the distance term entirely. Second, the ablation reported in the validation dossier found that reversing or dropping that term saves roughly a third of the path length with no measurable coverage penalty. The two studies used different environments, different metrics and different statistical tests, and reached the same conclusion.

The mechanism is straightforward once stated. The rule maximises its aggregate score, and the distance term enters as $+w_d \cdot \text{norm}(d)$. A frontier is therefore scored more highly the **further away** it is. Under equal weights this pull is strong enough to dominate the adjacency and trapezoid terms, and the robot systematically crosses explored ground to reach distant frontiers while nearer ones remain unvisited. Coverage still arrives eventually, which is why the defect is invisible to a coverage-only evaluation; the cost is paid entirely in travel.

Correcting the weight recovers most of the loss. TAD-tuned attains rank 3.96 against nearest-frontier's 3.58 — a gap well inside the critical difference of 1.36, so the tuned variant is competitive with the classical baseline. It does not, on this evidence, surpass it.

6.4 What the comparison says about the alternatives

Policy	Mean rank (AUC)	Standing
CU-exp	2.98	Best overall. The classical cost–utility trade-off remains hard to beat
CU-lin	3.06	Statistically tied with CU-exp; the two agreed on 38 of 60 trials
NF	3.58	Best on distance and on turning points. Remarkably strong for a 1997 rule with no parameters
TAD-tuned	3.96	Competitive once the distance weight is removed; not distinguishable from NF or LF
LF	4.00	Sound and parameter-free; frontier extent is a serviceable gain proxy
MIG	4.67	Pure information greed overreaches: it ignores travel cost and pays for it
TAD	6.84	Default weights; not separable from the random control

Policy	Mean rank (AUC)	Standing
Random	6.91	The floor against which the others are measured

A result worth dwelling on is how well nearest-frontier performs. It has no parameters, requires no tuning, and is the cheapest rule to evaluate, yet it leads on distance and turning points and sits within the critical difference of the best policy on efficiency. Any proposed replacement should be required to beat it convincingly before it is adopted, and on this evidence TAD at default weights does not.

7. Threats to Validity and Scope of the Claim

What this study does and does not claim

It evaluates **this project's grid-adapted implementation** of TAD, with min–max normalisation and equal default weights, in a kinematic simulator on reconstructed indoor environments. It is **not** a refutation of the published algorithm. The published formulation combines its terms in commensurable Rmap-rectangle units without normalisation, and the normalisation this project adds — necessary because its grid-adapted terms carry different units — is itself a plausible origin of the imbalance measured here. Establishing what the original formulation does would require implementing the Rmap representation, which this project does not build.

Threat	Mitigation	Residual risk
Policies might differ in more than their selection rule	Map, pose, sensor, motion, detection and termination are shared code, not matched settings	Low
Our method could be favoured by tuning	Competitors tuned on the same held-out seeds with the same budget and objective; the tuner chose against our default	Low
Results might be an artefact of one environment	Three environment families, 20 seeds each, blocked design; the ranking is consistent across all three panels of Figure 1	Low
Metric choice could favour a particular policy	Four metrics reported; TAD ranks last or next-to-last on three of them	Low
Kinematic simulation omits physics, Nav2 and real SLAM noise	Stated as scope; all policies share the same idealisation	Open — ranking may shift on hardware
Reconstructed environments are not the published ones	Comparison is internal (policy vs policy), not against published figures	Open

8. Conclusions and Recommendations

#	Conclusion	Evidence
1	All eight policies complete the exploration task to approximately 98% coverage; coverage alone cannot discriminate between them	Friedman on coverage, 60 paired trials
2	TAD at default weights is not statistically separable from random frontier selection on exploration efficiency	Mean ranks 6.84 vs 6.91, CD = 1.36
3	TAD at default weights is the least economical policy tested, travelling about 86% further than nearest-frontier for equal coverage	118 m vs 63 m
4	The deficiency is attributable to the polarity and magnitude of the distance term, not to the multi-criteria structure	Independent tuner selected $w_d = 0.0$; ablation agrees
5	With the distance weight removed, TAD becomes competitive with the classical baselines but does not surpass them	Ranks 3.96 vs NF 3.58, within CD

8.1 Recommended actions

Priority	Action	Cost
1	Change the default launch weights to $w_{\text{distance}} = 0$ (retaining adjacency and trapezoid). This is a parameter change, not a code change, and recovers most of the measured shortfall	None — already exposed at launch
2	Retain nearest-frontier as the reference configuration for regression testing; any future change should be required to beat it	None
3	If the distance criterion is to be retained with meaning, reconsider its sign so that proximity is rewarded, and re-tune	Small; one line plus a tuning sweep
4	Investigate whether the min–max normalisation, introduced to reconcile incommensurable units, is itself responsible for the imbalance	Moderate; requires the Rmap representation

Overall assessment

The TAD implementation is correct, as established by the validation dossier, but at its shipped default weights it is not an effective exploration policy: on this evidence it performs no better than random frontier selection and costs the most distance of any policy tested. The defect is narrow, well-localised and cheap to remedy — a single launch parameter recovers most of the gap. That a correctly implemented algorithm can still be the wrong thing to run is the central lesson of this comparison, and it is a conclusion that conformance testing alone could never have reached.

8.2 Reproducing this study

```
cd comparison
python3 benchmark.py # tuning + evaluation -> results_comparison.json
python3 analyze.py # Friedman/Nemenyi + figures
python3 build_comparison_report.py
```

The evaluation comprised 480 scored episodes plus the tuning sweep, 33 minutes of compute in total. `results_comparison.json` retains the full coverage-versus-distance trace of every episode, so all curves, statistics and figures can be recomputed independently.